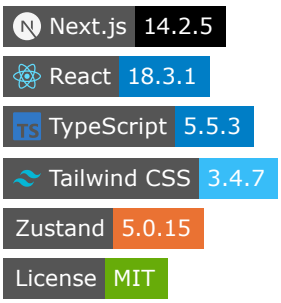
🎓 SkillBridge — Smart Internship & Job Readiness Platform

> ***Smart India Hackathon (SIH) Prototype***
> *Bridging the gap between students and industry through AI-driven ATS scoring, Career Twin evidence verification, and adaptive job readiness indexing.*



📋 Table of Contents

- Executive Summary & Problem Statement
- System Architecture & Data Flow
- Complete Feature Breakdown
- 1. Multi-Format Resume Intelligence & ATS Scorer
- 2. The Career Twin & Evidence Verification Matrix
- 3. Job Readiness Index (JRI) Mathematical Engine
- 4. Semantic Search & Weighted Skill Matching
- 5. Timed Adaptive Multi-Discipline Assessment Engine
- 6. Personalized Milestone Roadmap & Resource Curation
- 7. Skill Quest: Gamified RPG-HUD & Badge Locker
- 8. Dual-Criteria Internship Discovery & Why-Matched Breakdown
- 9. Kanban Application Lifecycle Tracker
- 10. Career Simulator ("What If You Pivoted?")
- 11. Live JD Analyzer & AI Resume Bullet Generator
- 12. Verified Digital Portfolio & One-Click PDF Generator
- UI & Route Sitemap
- Algorithmic Specifications & Mathematical Models
- Tech Stack & Library Justifications
- Data Models & Schema Reference
- Local Installation & Setup

- Contributing & Roadmap

🌟 Executive Summary & Problem Statement

The Problem

In today's collegiate landscape, over **80% of engineering students struggle to secure internships**, while tech recruiters face an influx of hundreds of generic resumes per opening:
1. **The "Black Box" ATS Trap**: Students submit PDF/DOCX resumes without knowing how enterprise Applicant Tracking Systems (ATS) evaluate their formatting, keywords, or structure.
2. **Unverified Claims**: Traditional portals let applicants check boxes for skills without verification, forcing recruiters to rely on crude filters (college tier or GPA cutoffs).
3. **Missing Feedback Loops**: Rejected applicants never learn *why* they were disqualified or what actionable skills they need to learn next.
4. **Mismatched Roles**: Students are often uncertain which career path aligns with their current skills, or how pivoting (e.g. from Frontend to Java Backend) affects their employability.

The SkillBridge Solution

SkillBridge is a comprehensive, student-first career acceleration platform developed for the **Smart India Hackathon**. It bridges students and recruiters through a transparent, evidence-based paradigm:
- **Instant ATS Transparency**: In-browser parsing of PDF, DOCX, and TXT resumes across a 7-dimensional scoring rubric.
- **Evidence-Backed Career Twin**: Skills are verified across multiple channels (Resume + Projects + Adaptive Assessments), transforming self-proclaimed claims into trusted proof.
- **Explainable Job Readiness Index (JRI)**: A multi-factor mathematical score that tells the student exactly where they stand and calculates projected gains for every action taken.
- **Closed-Loop Action Pathway**: Instead of dead-end rejections, SkillBridge pinpoints the student's `#1 Top Blocker`, prescribes curated open-access resources, and offers instant re-testing to verify mastery.

🏗️ System Architecture & Data Flow



🚀 Complete Feature Breakdown

1. Multi-Format Resume Intelligence & ATS Scorer

- **Zero-Dependency Native Parsing (`src/lib/resume-parser.ts`)**:
- **PDF Documents**: Direct parsing of binary stream buffers, extraction of `Tj` text operators, octal character decode, and whitespace normalization without requiring server-side Python or poppler dependencies.
- **Word DOCX Documents**: ZIP-container inspection targeting `word/document.xml` to parse `<w:t>` XML tokens.
- **Text & Markdown**: Full UTF-8 extraction and newline preservation.
- **Intelligent Entity Recognizers**:
- Contact information detection (RFC-5322 email regex, international phone numbers, profile links to GitHub, LinkedIn, Behance, Dribbble, Figma).
- Education history parser detecting degree acronyms (`B.Tech` , `BCA` , `M.Tech` , `B.Sc` , `MBA`), branch/field, and graduation year ranges.
- Experience & project extraction with company names, roles, date intervals, and technology badges.
- Impact verb classifier (`built` , `optimized` , `reduced` , `shipped` , `architected` , `automated`) and numerical metric quantification counter (`35%` , `2x` , `12k MAU` , `180ms`).
- **7-Dimension Enterprise ATS Scoring (`src/lib/ats-scorer.ts`)**:

- **Skills Density** (20%): Evaluates breadth and relevance against target roles.
- **Experience Quality** (20%): Evaluates role count, bullet count, and tenure.
- **Contact Information** (15%): Verifies presence of name, phone, email, and external links.
- **Education Depth** (15%): Verifies institution, degree, field, and year.
- **Projects Quality** (10%): Scans project portfolio and tech stack diversity.
- **Formatting & Structure** (10%): Flags resumes that are too short (<180 words), too long (>1200 words), or lack proper section breaks.
- **Impact Quantification** (10%): Measures use of hard metrics and strong action verbs.
- **Actionable ATS Feedback & Fix Cards**:
- Produces dynamic `strengths` and `improvements`.
- Provides pre-formatted **"Before vs. After"** bullet transformations that students can copy to their clipboard in one click.

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2. The Career Twin & Evidence Verification Matrix

SkillBridge rejects the notion of "self-proclaimed" skills. Instead, every skill in the candidate's profile is tracked via an evidence-backed tri-state model:

```
[ UNVERIFIED (0.2) ] -----> [ PARTIAL (0.6) ] -----> [ VERIFIED (0.95) ]
  (Self-claimed /            (Found on Resume OR      (Found in >=2 sources OR
   Target Role Gap)           Verified Project)        Scored >=80% in Assessment)
```

- **Verification Sources**:
- `RESUME`: Extracted from raw resume text or technical skills section.
- `PROJECT`: Evidenced in a parsed or submitted project tech stack.
- `ASSESSMENT`: Proved by scoring >= 80% on tagged domain questions in the timed assessment.
- `SELF`: Manually added or selected during onboarding.
- **Interactive Gap Matrix (`src/app/student/skills/page.tsx`)**:
- Compares candidate skills directly against industry target roles (e.g. *Full-Stack Developer*, *Java Backend Developer*, *Data Scientist*).
- Sorts gaps by priority: `Critical` (High-weight requirement unfulfilled), `High`, and `Medium`.
- Highlights the **#1 Top Roadblock** ("*Your biggest roadblock is Next.js for Full-Stack Developer*") and routes the student directly to verification.

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3. Job Readiness Index (JRI) Mathematical Engine

The **Job Readiness Index (JRI)** is a holistic score (0–100) reflecting a student's real-world employability:

$$\text{JRI} = \text{round}\Big(0.20 \cdot \text{Resume} + 0.25 \cdot \text{Technical} + 0.25 \cdot \text{Assessment} + 0.15 \cdot \text{Projects} + 0.10 \cdot \text{Aptitude} + 0.05 \cdot \text{Profile}\Big)$$

Weight Distribution

Dimension	Weight	Description
Technical Skills	**25%**	Percentage of verified skills (Verified = 100%, Partial = 50%)
Assessment Score	**25%**	Score achieved in the comprehensive 90-minute assessment
Resume ATS Health	**20%**	Overall 7-dimension ATS benchmark score
Projects Score	**15%**	Scaled project portfolio (25 pts per project, up to 100 pts)
Aptitude Score	**10%**	Average performance in Quantitative and Verbal reasoning
Profile Completeness	**5%**	Contact details, college, branch, CGPA, and social links

Readiness Bands

- **Top Candidate** (>= 85): Ready for premier product companies and high-stipend roles.
- **Strong** (70 - 84): Solid candidate meeting all baseline engineering requirements.
- **Ready** (55 - 69): Eligible for entry-level internships with minor skill polishing.
- **Developing** (35 - 54): Foundational knowledge present; requires structured project work.
- **Beginner** (< 35): Early stage; needs onboarding and core fundamentals.

Projected Gains Engine

Explains *why* the student has their current score and calculates dynamic "what-if" score increases:
- `+3 pts` if you verify Docker
- `+4 pts` if you add 1 more project to your resume
- `+7 pts` if you score >= 80% in the assessment

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4. Semantic Search & Weighted Skill Matching

- **TF-IDF Vector Space Retrieval (`src/lib/semantic-search.ts`)**:
- Indexes all internship postings across titles, locations, descriptions, and required stacks.
- Expands candidate queries with resume tokens, project descriptions, detected skills, and domain aliases.
- Employs sub-linear term frequency scaling with Euclidean document length normalization:
  $$\text{Score}(q, d) = \frac{1}{\sqrt{|d|}} \sum_{t \in q \cap d} \text{IDF}(t) \cdot (1 + \ln(\text{TF}_{t, d}))$$
- Returns ranked hits with semantic explanations: *"matches your skills: React, TypeScript; shared context: dashboard, B2B"*.
- **Weighted Skill Matching (`src/lib/skill-matcher.ts`)**:
- Compares candidate skills against job requirements weighted by importance (1 - 10).
- Partial credit (50%) awarded for contextual matches in experience bullet points.

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5. Timed Adaptive Multi-Discipline Assessment Engine

- **90-Minute Timed Test Environment (`src/app/student/assessment/page.tsx`)**:
- Full timer with visual warnings, question palette, and section switcher.
- Real-time status palette tracking:
  - 🟢 **Answered**
  - 🟣 **Marked for Review**
  - ⚪ **Unanswered**
  - 🔵 **Current Question**
- **4 Comprehensive Question Sections (`src/mock/questions.ts`)**:
- **Technical**: Role-specific engineering across Web, Java, Python/Data Science, Mobile, and UI/UX.
- **DSA**: Data structures, Big-O complexity, binary search, tree traversals, Dijkstra, and sorting algorithms.
- **Verbal Ability**: Syllogisms, coding-decoding, reading comprehension, analogies, and grammatical correction.
- **Quantitative Aptitude**: Time-speed-distance, probability, compound interest, permutations, and algebra.
- **Question Typologies**:
- Conceptual Multiple Choice (MCQ)
- Code Output Prediction
- Code Debugging & Syntax Bug Identification
- **Automated Skill Promotion**:
- Every question is tagged with a `skillTag` (e.g., `react`, `sql`, `spring`, `dsa`, `stats`).
- Upon submission, the engine evaluates accuracy per skill tag. Scoring >= 80% on tagged questions automatically promotes that skill to `VERIFIED` status in the student's Career Twin!

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6. Personalized Milestone Roadmap & Resource Curation

- **5-Stage Milestone Pathway (`src/app/student/roadmap/page.tsx`)**:
$$\text{Gap Identified} \longrightarrow \text{Learn Module} \longrightarrow \text{Practice Task} \longrightarrow \text{Build Project} \longrightarrow \text{Re-assess \& Verify}$$
- **Curated Open-Access Resource Hub (`src/mock/resources.ts`)**:
- Directly maps identified skill gaps to curated tutorials from **freeCodeCamp**, **MDN Web Docs**, **NPTEL**, **LeetCode**, **GeeksforGeeks**, and **Official Docs**.
- Displays estimated duration (e.g. `2h`, `6h`, `40h`) and difficulty levels (`Beginner`, `Intermediate`, `Advanced`).

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7. Skill Quest: Gamified RPG-HUD & Badge Locker

- **Experience Points (XP) & Progression (`src/app/student/skill-quest/page.tsx`)**:
- Level formula: $\text{Level} = 1 + \lfloor \frac{\text{XP}}{500} \rfloor$.
- Students earn XP for uploading resumes (+50 XP), taking assessments (+100 XP), and verifying skills.

- **Daily Streak Tracker**:
- Promotes consistent learning through activity streaks with animated fire indicators.
- **Mission Board Quests**:
- *Resume Master*: Score >= 80% on ATS scoring (+50 XP).
- *Aptitude Ace*: Clear the Quantitative section (+100 XP).
- *Full-Stack Artisan*: Verify 3 backend skills (+250 XP).
- *Streak Keeper*: Maintain a 3-day streak (+30 XP).
- *Project Builder*: List 3+ verified projects (+80 XP).
- **Unlockable Badges**:
- 📄 Resume Master
- 🧮 Quant Ace
- ⚡ DSA Decimator
- 🛠️ Verifier
- 🔥 Streak Keeper

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8. Dual-Criteria Internship Discovery & Why-Matched Breakdown

- **Fixed vs. Improvable Eligibility Engine (`src/app/student/internships/[id]/page.tsx`)**:
- **Fixed Criteria (Hard Pass/Fail)**:
  - Minimum CGPA cutoff (e.g. >= 6.5)
  - Eligible branches (e.g. `CSE` , `IT` , `ECE`)
  - Graduation year eligibility (e.g. `2026` , `2027` , `2028`)
  - *If fixed criteria fail, the application is disabled with clear explanations.*
- **Improvable Criteria (Actionable Gaps)**:
  - Specific skills required by the company.
  - Direct links to *"Open Quest"* or *"Take Assessment"* to turn a rejection into an interview.
- **"Why This Match?" Explainability Modal (`src/app/student/internships/page.tsx`)**:
- Transparent point-by-point breakdown of the candidate's match percentage:
  - Skills Fit (50 pts)
  - Education Fit (20 pts)
  - Experience Fit (10 pts)
  - Assessment Fit (15 pts)

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9. Kanban Application Lifecycle Tracker

- **Stage Management (`src/app/student/applications/page.tsx`)**:
- Full lifecycle tracking across 6 distinct pipeline columns:
  1. `Applied`

2. `Shortlisted`
3. `Online Assessment`
4. `Technical Interview`
5. `Selected`
6. `Rejected`

- **Application History & Interview Notes**:
- Automatically records timestamped audit logs whenever a candidate advances through a stage.
- Integrated private note-taking area for interview preparation and HR contact details.

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10. Career Simulator ("What If You Pivoted?")

- **Instant Role Pivot Sandbox (** `src/app/student/career-simulator/page.tsx` **)**:
- Lets students simulate switching their career target across 10 industry roles:
  - Full-Stack Developer
  - Java Backend Developer
  - Python Developer
  - Data Analyst
  - Data Scientist
  - Cybersecurity Analyst
  - Cloud Engineer
  - DevOps Engineer
  - AI/ML Engineer
  - UI/UX Designer
- Recalculates JRI, identifies new top roadblocks, and highlights academic eligibility differences in real time without overwriting their primary profile.

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11. Live JD Analyzer & AI Resume Bullet Generator

- **Job Description Semantic Matching (** `src/app/student/job-analyzer/page.tsx` **)**:
- Paste any external job description to generate an instant semantic compatibility score.
- Highlights exact keyword hits and identifies critical missing requirements.
- **Impact Bullet Generator**:
- Provides pre-engineered, metric-driven resume bullet points tailored to the specific role that students can copy directly into their resume.

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12. Verified Digital Portfolio & One-Click PDF Generator

- **Shareable Public Profile (** `src/app/student/portfolio/page.tsx` **)**:

- Toggle between **Public** and **Private** visibility.
- Generates a shareable vanity link: `https://skillbridge.local/p/student-name`.
- Displays verified skills with badges showing proof sources (e.g. *Verified by RESUME + ASSESSMENT*).
- **One-Click Print & PDF Generator**:
- Generates a clean, ATS-formatted printable HTML resume view styled specifically for browser printing or saving as PDF.

🗺️ UI & Route Sitemap

```
/                           Landing page with platform overview & quick start
/auth
  /login                    Student login via academic email
  /signup                   Career Twin creation (College, Branch, CGPA, Grad Yea
/onboarding                 4-Step Progressive Onboarding Flow
  /profile                  Step 1: Academic & personal credentials
  /interests                Step 2: Technology domain selection
  /roles                    Step 3: Target role selection (up to 2 roles)
  /resume                   Step 4: Drag-and-drop resume upload & parsing
/student                    Authenticated Student Portal (Shell Layout)
  /dashboard                Central command center (JRI dial, top blocker, active
  /resume-analysis          7-dimension ATS analysis, scores & before/after fixes
  /skills                   Evidence-based skill matrix & gap analyzer
  /assessment               90-minute timed adaptive assessment with live palette
    /result                 Assessment scorecard & promoted verified skills
  /roadmap                  Milestone pathway with curated learning resources
  /skill-quest              RPG-style HUD, XP progression, streaks & badges
  /internships              Searchable internship catalog with match percentages
    /[id]                   Detailed internship spec with Pass/Fail fixed criteri
  /applications             Kanban pipeline application tracker
  /portfolio                Digital verified portfolio with PDF export & public l
  /career-simulator         "What if you pivoted?" real-time role sandbox
  /job-analyzer             Paste external JD for semantic match & bullet suggest
/upload                     Standalone resume upload portal
  /results                  Standalone ATS analysis & semantic search results
/api
  /profile                  REST endpoint for server-side profile persistence
```

🧮 Algorithmic Specifications & Mathematical Models

1. Job Readiness Index (JRI)

```
const JRI = Math.round(
  0.20 * resumeAtsScore +
  0.25 * technicalSkillsScore +
  0.25 * assessmentScore +
  0.15 * projectsScore +
  0.10 * aptitudeScore +
  0.05 * profileCompleteness
);
```

- **Technical Skills Score**: `(Verified * 100 + Partial * 50) / Total Skills`
- **Projects Score**: `min(100, Project Count * 25)`

- **Aptitude Score**: `(Quant Score + Verbal Score) / 2`
- **Profile Completeness**: Percentage of 6 core profile attributes present.

2. Weighted Internship Match Algorithm

```
const totalWeight = requiredSkillIds.reduce((acc, id) => acc + skillWeight(id), 0);
let earnedWeight = 0;

for (const id of requiredSkillIds) {
  if (candidateHasSkill(id)) {
    earnedWeight += skillWeight(id);
  } else if (candidateHasPartialSkill(id)) {
    earnedWeight += skillWeight(id) * 0.5; // 50% partial credit
  }
}

const matchPct = Math.round((earnedWeight / totalWeight) * 100);
```

3. Semantic TF-IDF Search Engine

- Token extraction via pattern `/[a-z0-9+#.]+/g` with stopword filtration.
- Term Frequency (TF) and Inverse Document Frequency (IDF):
  $$\text{IDF}(t) = \ln\left(1 + \frac{N}{1 + \text{DF}(t)}\right)$$
  $$\text{Score} = \frac{1}{\sqrt{L_{\text{doc}}}} \sum_{t} \text{IDF}(t) \cdot (1 + \ln(\text{TF}_t))$$

💻 Tech Stack & Library Justifications

Layer	Technology	Version	Justification
Framework	**Next.js**	`14.2.5`	React Server Components, App Router, and fast client-side transitions.
Language	**TypeScript**	`5.5.3`	Type-safety across candidate profiles, assessments, and internship schemas.
Styling	**Tailwind CSS**	`3.4.7`	Custom design system (`brand` , `ink`), responsive utility classes, micro-animations.
Client State	**Zustand**	`5.0.15`	Lightweight state management with LocalStorage persistence middleware.
Icons	**Lucide React**	`0.439.0`	Accessible, modern icon set.
File Persistence	**Node.js** `fs`	Built-in	Atomic file-writing queue for local profile storage (`.data/profile.json`).
Server Limit	**Server Actions**	Next Config	Configured `bodySizeLimit: 10mb` for handling rich multi-page PDF/DOCX resumes.

📊 Data Models & Schema Reference

CareerTwin

```ts
interface CareerTwin {
  profile: StudentProfile | null;
  ats: AtsResult | null;
  resume: ParsedResume | null;
  skills: SkillEvidence[];
  targetRoleIds: TargetRoleId[];
  interests: string[];
  jri: JobReadinessIndex;
  assessment: {
    lastScore: number;
    lastTechScore: number;
    lastVerbalScore: number;
    lastQuantScore: number;
    lastDsaScore: number;
    completedAt?: string;
  };
  xp: number;
  level: number;
  streak: number;
  applications: ApplicationRecord[];
  portfolioPublic: boolean;
}
```

SkillEvidence

```ts
interface SkillEvidence {
  skillId: string;
  skillName: string;
  status: "VERIFIED" | "PARTIAL" | "UNVERIFIED";
  confidence: number;
  sources: Array<"RESUME" | "PROJECT" | "ASSESSMENT" | "SELF">;
  assessmentScore?: number;
}
```

InternshipOpportunity

```ts
interface InternshipOpportunity {
  id: string;
  title: string;
  company: string;
  logoColor: string;
  location: string;
  stipend: string;
  durationWeeks: number;
  domain: string;
  description: string;
  requiredSkillIds: string[];
  niceToHaveSkillIds?: string[];
  fixedCriteria: {
    minCgpa: number;
    branch: string[];
    gradYear: number[];
  };
  improvableCriteria: string[];
  applicants: number;
  postedAt: string;
}
```

⚡ Local Installation & Setup

Prerequisites

- **Node.js**: v18.17.0 or higher
- **npm**: v9.0.0 or higher (or `pnpm` / `yarn`)

Quick Start

```
# 1. Clone the repository
git clone https://github.com/piyushverma-1/sih.git
cd sih

# 2. Install dependencies
npm install

# 3. Start the Next.js development server
npm run dev
```

Open http://localhost:3000 in your browser to experience SkillBridge.

Production Build & Linting

```
# Run TypeScript compilation and production build
npm run build

# Start production server
npm run start

# Run ESLint validation
npm run lint
```

🗺 Contributing & Roadmap

Future Enhancements for Smart India Hackathon

- [ ] **AI-Powered Voice Mock Interviews**: Integrating browser speech-to-text with LLM interview feedback.
- [ ] **College Placement Cell Portal**: Admin dashboard for TPOs to view cohort readiness scores and batch analytics.
- [ ] **Direct Recruiter Dashboard**: Verified candidate search by verified skills and JRI thresholds.
- [ ] **Automated GitHub Repository Scraping**: Real-time project code validation and complexity analysis.

📄 License

This project is open-source software licensed under the MIT License. Built with ❤️ for the Smart India Hackathon.